

Q1) a) $x_1 = 5\sin(\pi 416t) + \sin(\pi 1144t) + 10\sin(\pi 1560t)$

Fundamental Frequency = $\text{GCD}(208, 572, 780)$
 $= 52 \text{ Hz}$

$\therefore \text{Time Period} = \frac{1}{52} \text{ s} = 0.0192 \text{ s}$ Ans

$x_2 = 20\sin(\pi 784t) + 5\sin(\pi 1568t) + 30\sin(\pi 2156t)$

Fundamental Frequency = $\text{GCD}(392, 784, 1078)$
 $= 98 \text{ Hz}$

$\therefore \text{Time Period} = \frac{1}{98} \text{ s} = 0.0102 \text{ s}$ Ans

b) Pitch is directly proportional to fundamental frequency

Higher the fundamental frequency $\rightarrow$ Higher the Pitch

F.F of $x_1 = 52 \text{ Hz}$ & $x_2 = 98 \text{ Hz}$

$\therefore$ They don't have the same pitch.
 Pitch of x_2 will be higher than x_1 .

Q3) 4th Harmonic : $A = 2, T = 0.2 \text{ s}$
 $f = \frac{1}{T} = \frac{1}{0.2} = 5 \text{ Hz}$

$\Rightarrow 2\sin(2\pi 5t)$

1st Harmonic : $A = 4, f = \frac{5}{4} = 1.25 \text{ Hz}$

$T = \frac{1}{f} = \frac{1}{1.25} = 0.8 \text{ s}$

$\Rightarrow 4\sin(2\pi 1.25t)$

2nd Harmonic : $A = 3, f = 1.25 \times 2 = 2.5 \text{ Hz}$

$T = \frac{1}{2.5} = 0.4 \text{ s}$

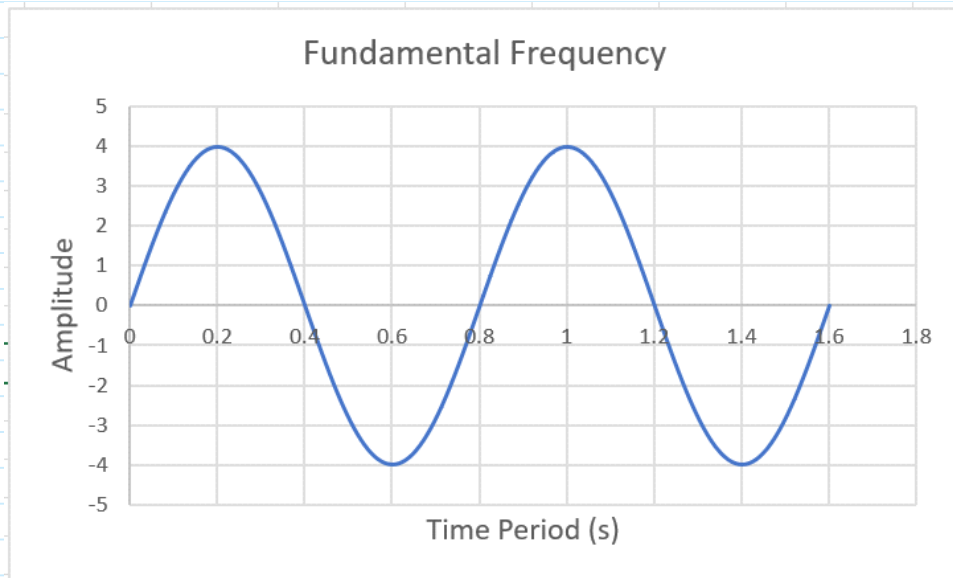
$\Rightarrow 3\sin(2\pi 2.5t)$

3rd Harmonic : $A = 2, f = 1.25 \times 3 = 3.75 \text{ Hz}$

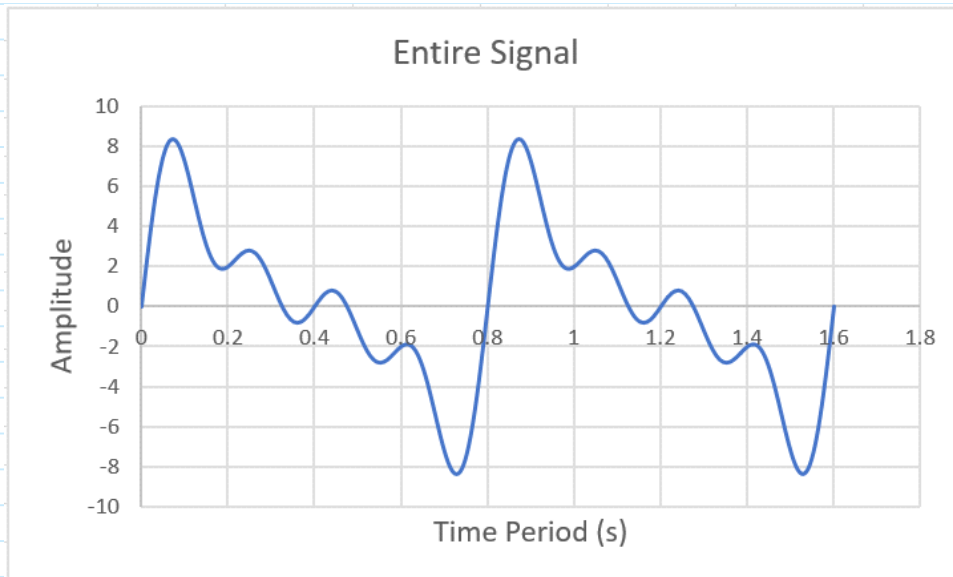
$T = \frac{1}{3.75} = 0.266 \text{ s}$

$\Rightarrow 2\sin(2\pi 3.75t)$

a)



b)



Q2)a $s(t) = 7\sin(2\pi 10000t) + 4\sin(2\pi 15000t) + 5\sin(2\pi 20000t) + 3\sin(2\pi 30000t) + 2\sin(2\pi 35000t)$

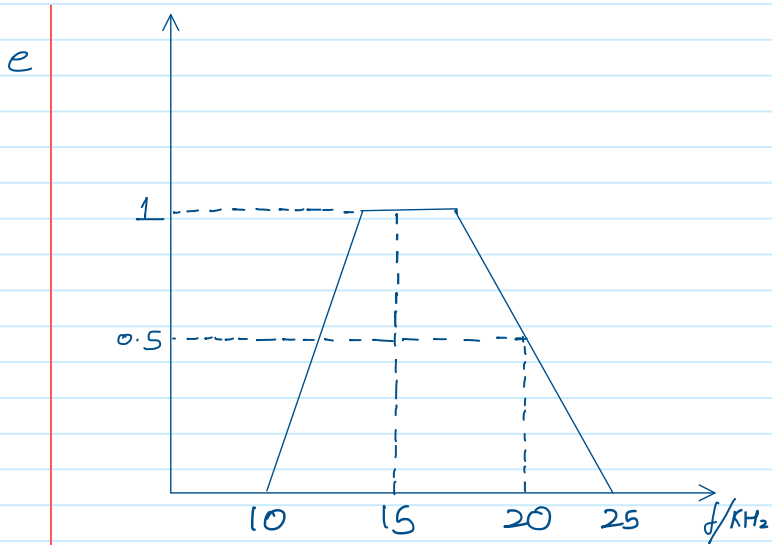
b fundamental frequency of $s(t)$:

GCD of (10, 15, 20, 30, 35)

$\Rightarrow$ 5 KHz Ans

c 3rd Harmonic, 4th Harmonic

d Bandpass Filter



Bandwidth of the filter = $25 - 10 = 15 \text{ KHz}$
Signal with frequency of more than 10 and less than 25 can pass through
Signal with frequency < 10 or > 25 will be blocked